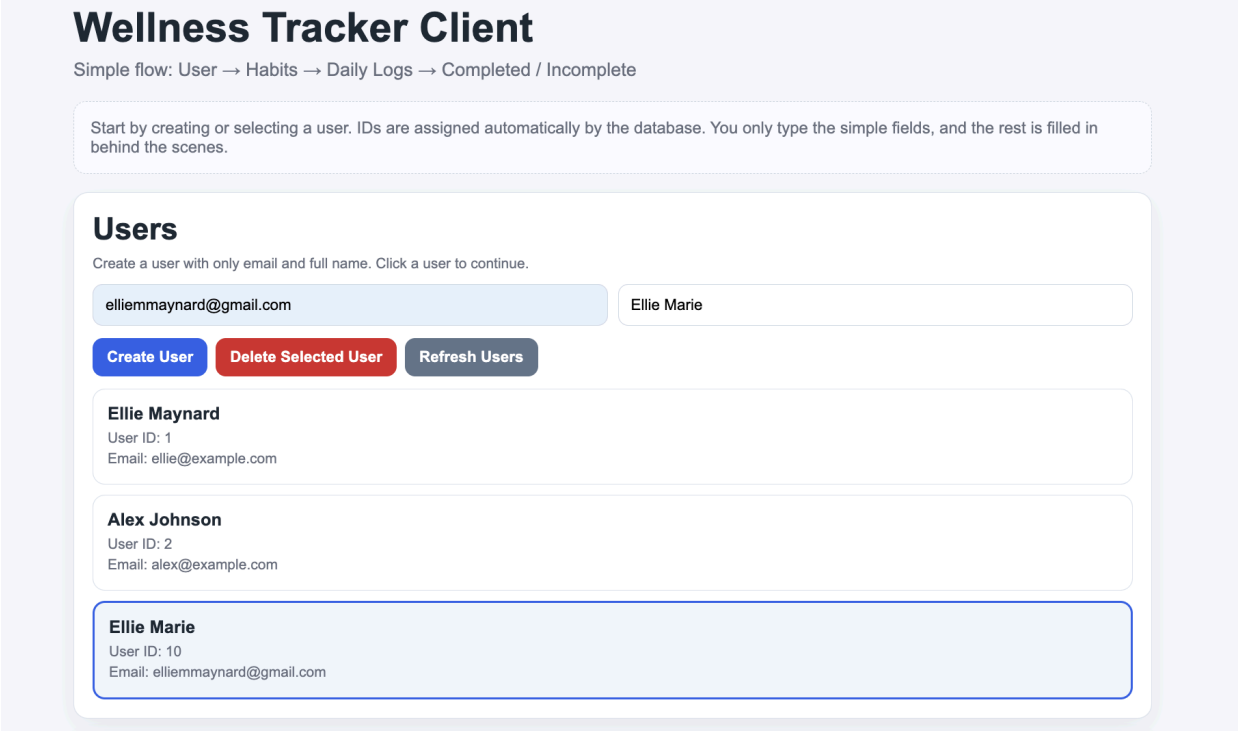
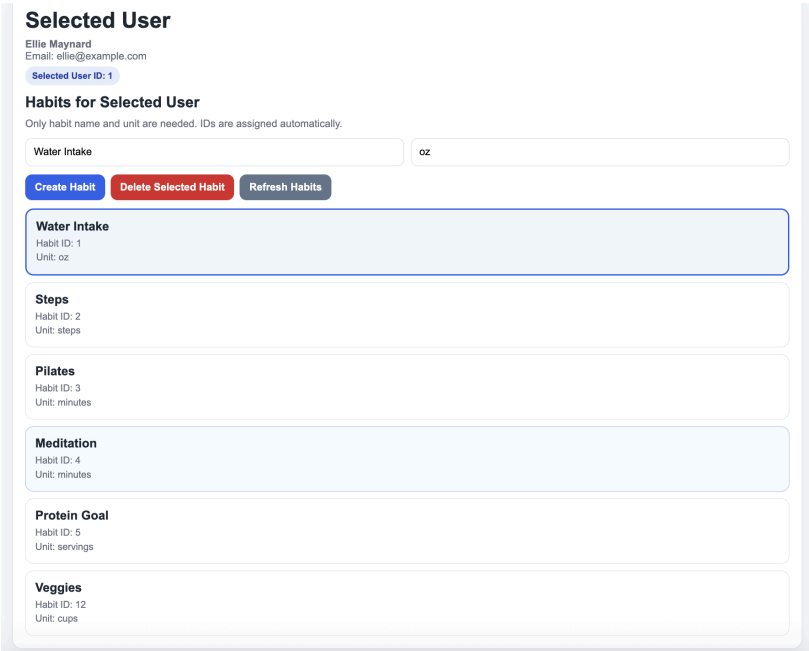
Create User:

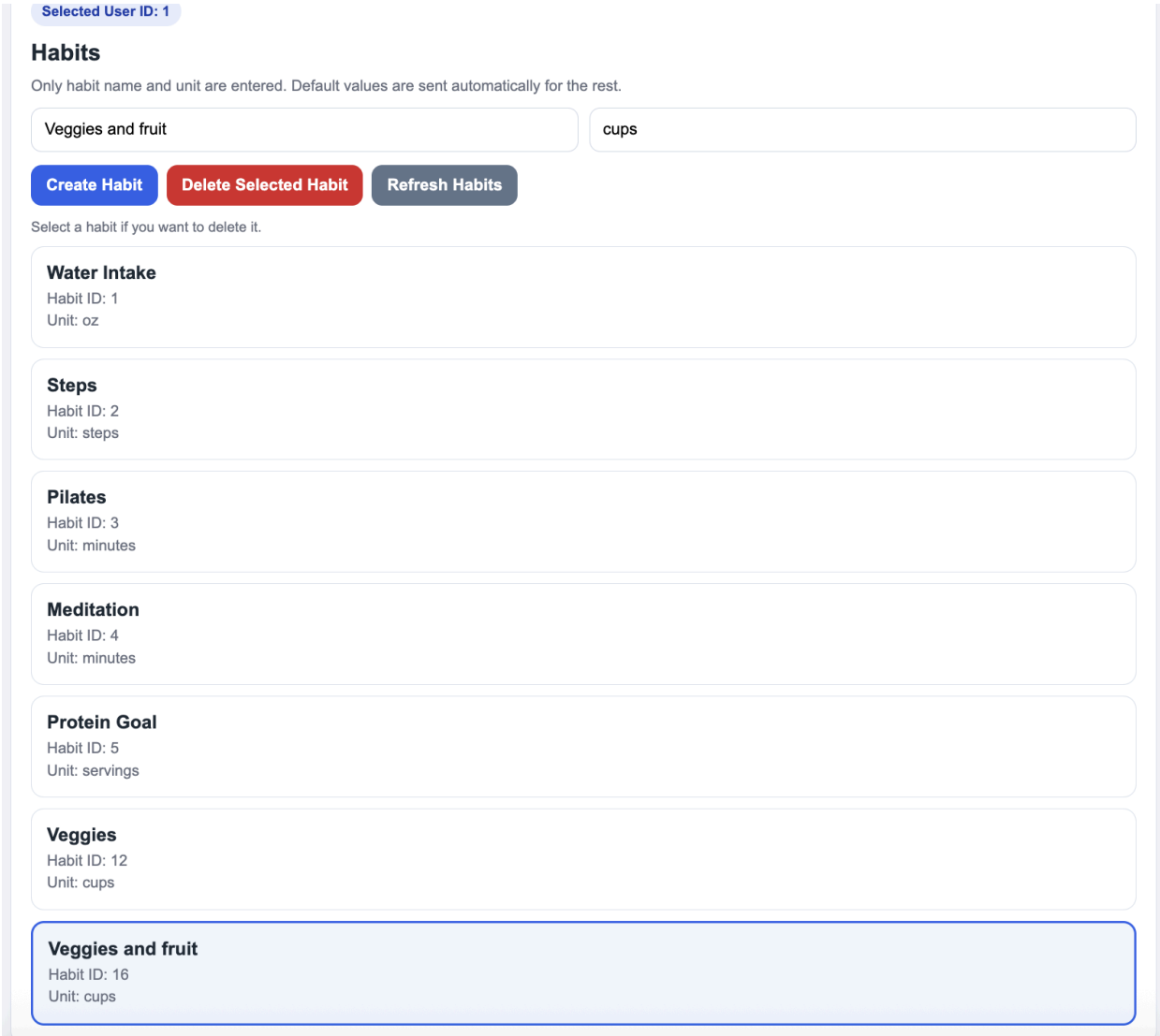



```
mysql> SELECT * FROM AppUser;
+---------+-------------------------+---------------+---------------------+
| user_id | email                   | full_name     | created_at          |
+---------+-------------------------+---------------+---------------------+
|       1 | ellie@example.com       | Ellie Maynard | 2026-02-23 16:10:33 |
|       2 | alex@example.com        | Alex Johnson  | 2026-02-23 16:10:33 |
|      10 | elliemmaynard@gmail.com | Ellie Marie   | 2026-03-16 22:22:03 |
+---------+-------------------------+---------------+---------------------+
3 rows in set (0.02 sec)
```

Create Habit:



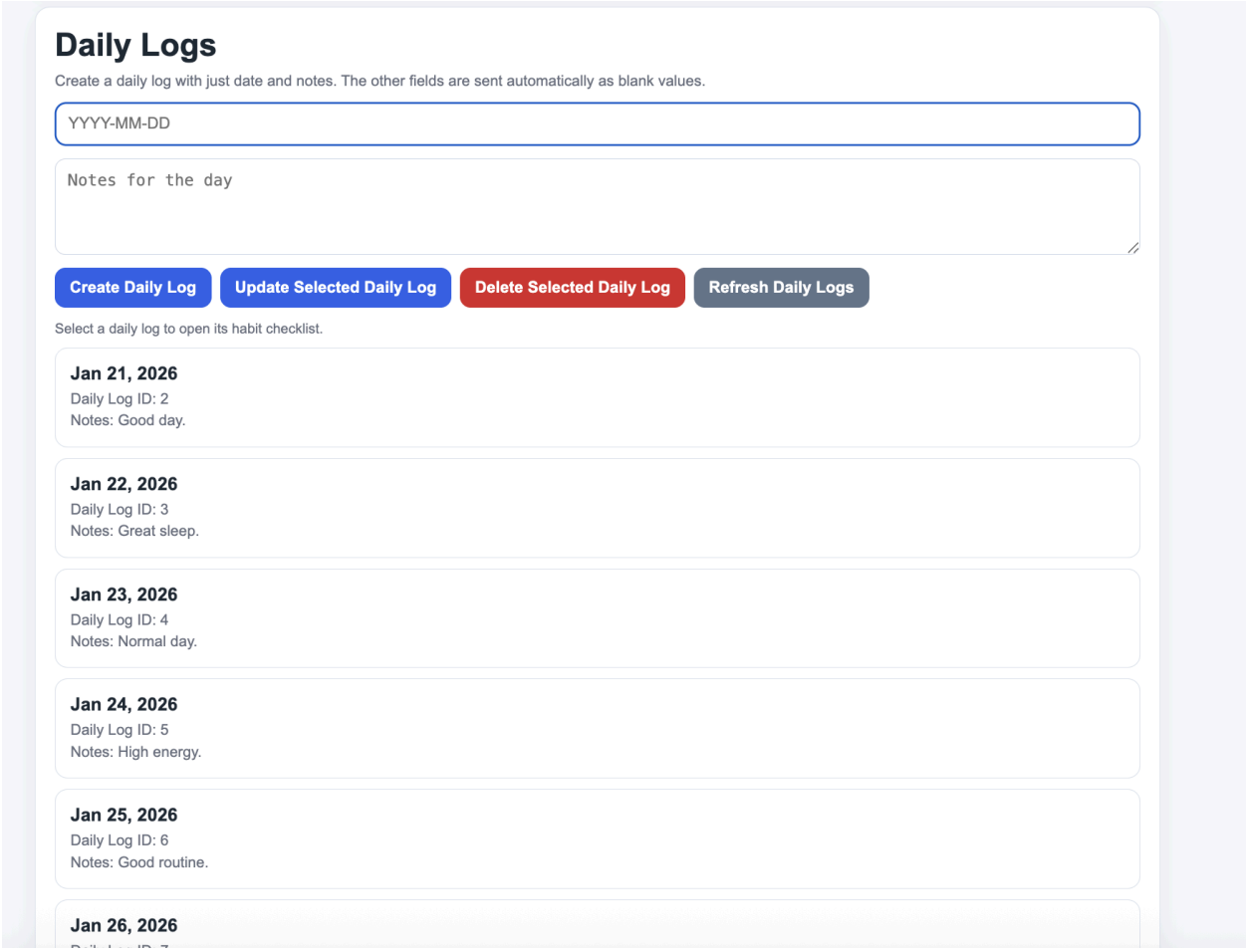
Create Habit/Get all habits for a User:



habit_id	user_id	name	unit	target_value	is_active
4	1	Meditation	minutes	10.00	1
3	1	Pilates	minutes	30.00	1
5	1	Protein Goal	servings	3.00	1
2	1	Steps	steps	10000.00	1
12	1	Veggies	cups	3.00	0
16	1	Veggies and fruit	cups	NULL	1
1	1	Water Intake	oz	3.00	1

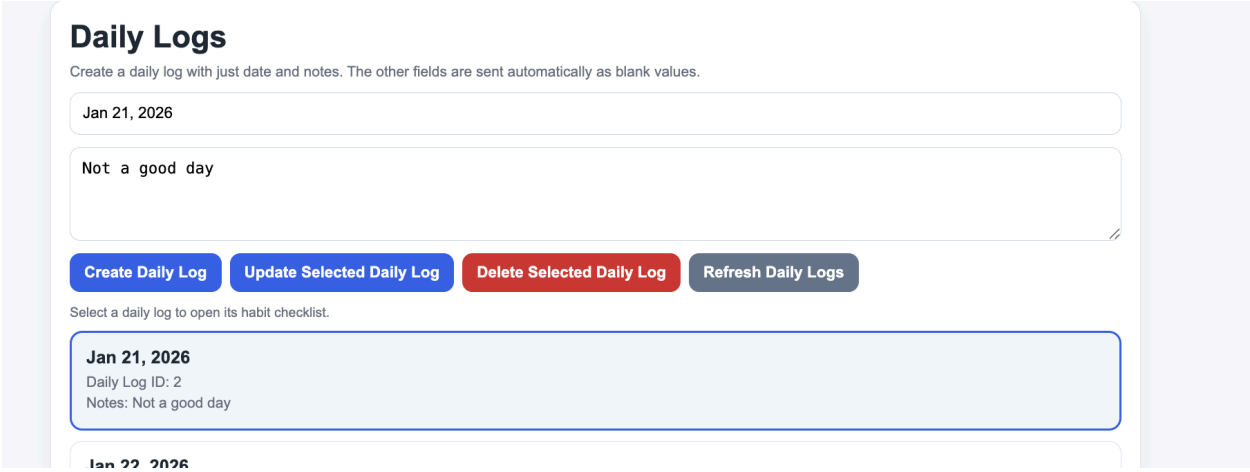
7 rows in set (0.41 sec)

Get all Logs:



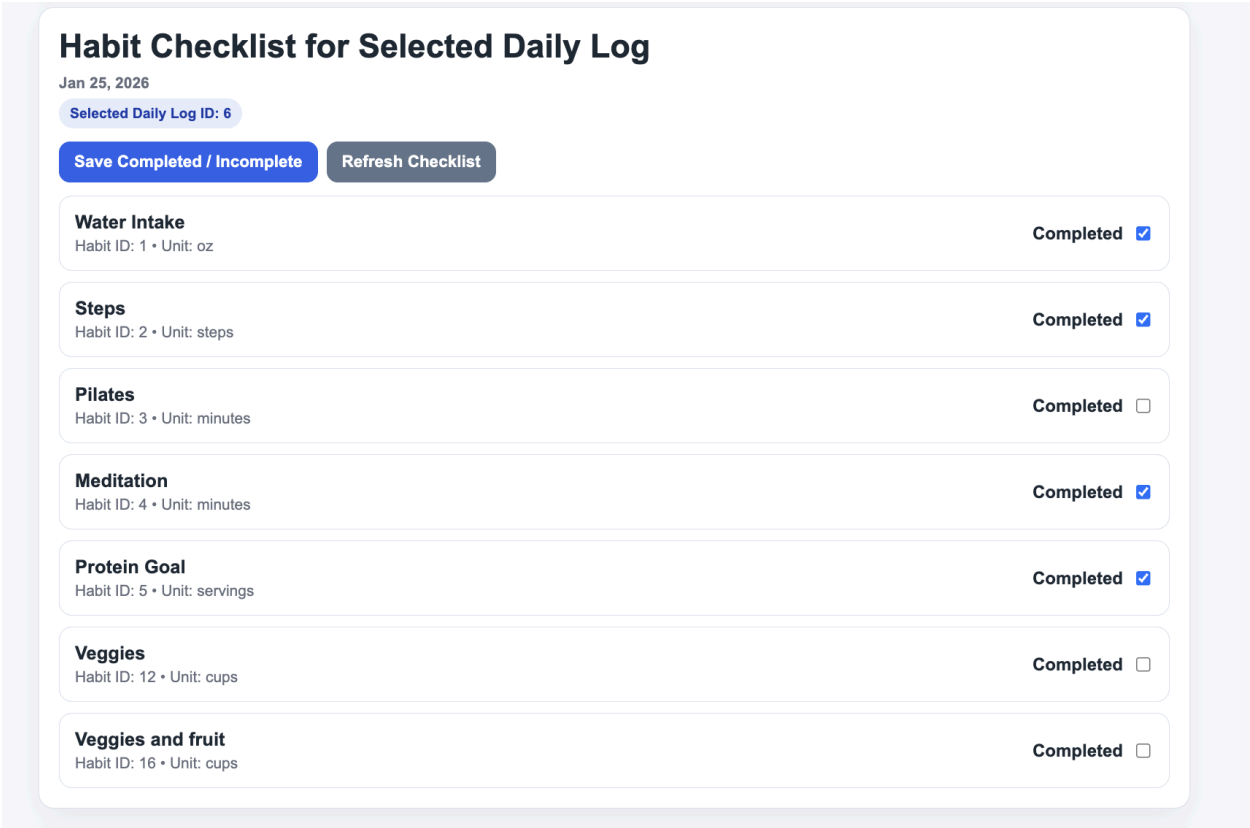

```
mysql> SELECT * FROM DailyLog
    -> WHERE user_id = 1;
+--------------+---------+------------+-------------+-------------+--------------+--------------+---------------
--+
| daily_log_id | user_id | log_date   | sleep_hours | mood_rating | stress_level | energy_level | notes
  |
+--------------+---------+------------+-------------+-------------+--------------+--------------+---------------
--+
|            2 |       1 | 2026-01-21 |        NULL |        NULL |         NULL |         NULL | Good day.
  |
|            3 |       1 | 2026-01-22 |        8.00 |           9 |            3 |            8 | Great sleep.
  |
|            4 |       1 | 2026-01-23 |        7.00 |           7 |            5 |            6 | Normal day.
  |
|            5 |       1 | 2026-01-24 |        8.50 |           9 |            2 |            9 | High energy.
  |
|            6 |       1 | 2026-01-25 |        7.20 |           8 |            4 |            7 | Good routine.
  |
|            7 |       1 | 2026-01-26 |        6.80 |           7 |            5 |            6 | Slightly tired
. |
|            8 |       1 | 2026-01-27 |        7.90 |           8 |            4 |            7 | Solid day.
  |
|            9 |       1 | 2026-01-28 |        6.50 |           6 |            7 |            5 | Stressful.
```

Update Log:




```
mysql> mysql> SELECT * FROM DailyLog WHERE user_id = 1;
+--------------+---------+------------+------------+-------------+--------------+-------------+--------------
--+
| daily_log_id | user_id | log_date   | sleep_hours | mood_rating | stress_level | energy_level | notes
  |
+--------------+---------+------------+------------+-------------+--------------+-------------+--------------
--+
|            2 |       1 | 2026-01-21 |       NULL |        NULL |         NULL |        NULL | Not a good day
  |
|            3 |       1 | 2026-01-22 |       8.00 |           9 |            3 |           8 | Great sleep.
```

Get habits by Daily Log:




```
mysql> SELECT h.*
    -> FROM Habit h
    -> JOIN HabitEntry e ON h.habit_id = e.habit_id
    -> WHERE e.daily_log_id = 6;
+----------+---------+--------------+----------+--------------+-----------+
| habit_id | user_id | name         | unit     | target_value | is_active |
+----------+---------+--------------+----------+--------------+-----------+
|        1 |       1 | Water Intake | oz       |         3.00 |         1 |
|        2 |       1 | Steps        | steps    |     10000.00 |         1 |
|        3 |       1 | Pilates      | minutes  |        30.00 |         1 |
|        4 |       1 | Meditation   | minutes  |        10.00 |         1 |
|        5 |       1 | Protein Goal | servings |         3.00 |         1 |
+----------+---------+--------------+----------+--------------+-----------+
5 rows in set (0.02 sec)
```

Deployment Document:

Wellness Tracker Deployment Guide

Overview

This document explains how to download, configure, build, run, and host the Wellness Tracker project from the GitHub repository. It is written for someone unfamiliar with the project who needs to go from downloading the source code to successfully running the application.

The project is a multi-layer Java application with:
- **Data layer**: MySQL database with DAO classes
- **Business layer**: manager/service classes
- **Service layer**: Javalin REST API
- **Client/front end**: HTML, CSS, and JavaScript client

The project is hosted with:
- **Back end/API**: Render
- **Database**: Railway MySQL
- **Front end/client**: GitHub Pages or local browser-hosted HTML

Project Components

Back End
The back end is a Java Maven project that includes:
- DAO classes
- business/manager classes
- model classes
- service/API layer
- JSON utility classes

The API is built with **Javalin** and packaged with **Maven**.

Front End
The front end is a lightweight HTML/JavaScript client that sends requests to the Javalin API.

Database
The database is MySQL and contains these main tables:
- `AppUser`
- `Habit`
- `DailyLog`
- `HabitEntry`

Prerequisites

Before running the project, install and configure the following:

Required Software
- **GitHub account**
- **Git**
- **Java JDK 17 or newer**
- **Apache Maven**
- **MySQL client or GUI**
  - MySQL Workbench, Railway query console, or MySQL CLI
- **IDE**
  - IntelliJ IDEA recommended
  - VS Code also works
- **Browser**
  - Chrome recommended
- **Render account**
- **Railway account**

Recommended Platforms

IDE
Use one of the following:
- IntelliJ IDEA
- VS Code with Java extensions

Database Platform
- Railway MySQL

Service Hosting Platform
- Render

Front End Hosting Platform
- GitHub Pages

Repository Download and Setup

Option 1: Download ZIP
1. Open the GitHub repository.
2. Click **Code**.
3. Click **Download ZIP**.
4. Unzip the project to your computer.
5. Open the folder in IntelliJ IDEA or VS Code.

Option 2: Clone Repository
```bash
git clone <your-repository-url>
cd <repository-folder>
```